BAHIR-DAR CITY ADMINISTRATION EDUCATION DEPARTMENT

BIOLOGY MODEL EXAMINATION FOR GRADE 12 2018 E.C

TOTAL NUMBER OF ITEMS -100 **TIME ALLOWED ----TWO HOURS**

INSTRUCTION;-CHOOSE YOUR BEST ANSWER FROM THE SUGGESTED OPTIONS AND WRITE THE LETTER OF YOUR CHOICE ON THE PROVIDED ANSWER SHEET.

B 1. Cleanising experiment in a way that enables to reduce suffering of animals using less invasive techniques, better medical care and living conditions is

A. Reduction B. Refinement C. Replacement D. Imagination

B 2. In which part of the oviduct of chicken fertilization takes place?

A. Magnum B. Infundibulum C. Isthmus D. Uterus

A 3. Which of the following is orientational animal behavior occurs either towards or away from the source of stimuli?

A. Taxis B. Kinesis C. Simple reflex D. Imprinting

C 4. Which one of the following operant behavior responses that neither increases nor decreases the probability of behavior being repeated?

A. Reinforces B. Punishers C. Neutral operants D. Habituation

C 5. Which part of the brain regulates body temperature?

A. Cerebellum B. Thalamus C. Hypothalamus D. Medulla

C 6. Which of the following hormones decreases blood sugar level?

A. Cortisol B. Adrenaline C. Insulin D. Glucagon

7. A certain insect usually becomes active each day at dusk. If kept in the dark all day, it will still become active around sun set even though it cannot see the sun; this behavior is an example of

A. Transverse orientation
C. Exogenous entrainment
B. Diurnal behavior
D. Circadian rhythm

8. An enzyme which is responsible for unwinding or unzipping of the two strands of DNA IS

A. Topoisomerase B. Helicase C. Ligase D. Polymerase

9. The sequences of amino acids in the polypeptide chain is called --------structure.

A. Primary B. Secondary C. Tertiary D. Quaternary

9. The sequence of amino acids in the polypeptide chain is called?

A. Primary Structure
C. Tertiary
B. Secondary
D. Quaternary

10. Which one of the following inhibitors binds only to the ES complex?

A. Competitive
C. non competitive
B. uncompetitive
D. Irreversible

11. Which class of enzyme is responsible for non-hydrolytic cleavage of bonds or transferring a certain specific group?

A. Ligase B. Hydrolase C. Lyase D. Transferase

12. Which one of the following enzymes is used for meat tenderization?

A. Rennin B. Papa in C. trypsin D. Ligninase

13. Which one of the following best describes the synthesis of the lagging strand?

A. Continuous synthesis in the 5'→3' direction
B. Discontinuous synthesis in the 3'→5' direction
C. Discontinuous synthesis in the 5'→3' direction
D. Continuous synthesis in3'→5' the direction

23. What is the primary purpose of creating transgenic organisms? To:

A. Generate pathogenic organisms that fight against non-pathogenic ones

B. Produce organisms that mimic human beings in every aspect of behavior

C. Promote expression of desired genes that improve nutritional quality of organism

D. Produce novel organisms that can affect reproduction among different species

A 24. One of the main concern research areas of professor Yalem Tsehay Mekonnen is?

A. using chemical pesticides on the hazard of human beings

B. to safeguard genetic diversity of Ethiopia

C. increase crop productivity by using genetic engineering

D. safeguard of biodiversity

D 25. The unique characteristics of fungi is?

A. All are multicellular and eukaryotic

B. only contains genetic material

C. can synthesize their own food

D. have cell wall made from chitin

photo organo hetero

B 26. The bacteria that use organic compounds as its source of carbon and light as its source of energy are?

A. photolithotrophs

B. photoorganoheterotrophs

C. chemolithoautotrophs

D. chemolithoheterotrophs

D 27. Which of the following bacterium is comma shaped?

A. Sarcina

B. Bacillus

C. palisade

D. Vibrio

B 28. Which one of the following viral replication cycles is correctly placed?

A. Penetration → Adsorption → Synthesis → Maturation → Release

B. Adsorption → Penetration → Synthesis → Maturation → Release

C. Adsorption → Penetration → Maturation → Synthesis → Release

D. Synthesis → Maturation → Release → Penetration → Adsorption

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A 14. The longest phase of interphase is?

A. G1 phase B. G2 phase C. S phase D. M phase

B 15. The stage in meiosis where by sister chromatids separate and migrate towards opposite poles is?

A. Anaphase I B. Anaphase II C. Prophase I D. Metaphase II

B 16. Which one of the following is not true about the genetic code?

A. Triplet B. Overlapping C. Degenerate D. Universal

B 17. If the triplet code is AGC in the antisense strand of DNA, what is the anti codon in tRNA?

A. UCG B. AGC C. TCG D. TGC

D 18. The joint that opens and closes in one direction?

A. Ball and socket B. Pivot C. Saddle D. Hinge

C 19. The only contraceptive method that prevents STIs is?

A. Diaphragm B. Cervical cap C. Condom D. IUD

A 20. Which one of the STIs is not caused by bacteria?

A. Genital warts B. Chancroid C. Gonorrhea D. Syphilis

C 21. 80 individuals are born per year in a population, and the midyear population is 5000. What is the birth rate of the area?

A. 20 B. 16 C. 4 D. 2

B 22. What is the role of Biology in the investigation of a criminal case? To:

A. Detect flavor compounds in food and monitor environmental pollutant

B. Examine potential source of evidence using blood, saliva, and hair

C. Sense pathogens and herbicides using an immunochemical system

D. Determine the production of complex organic substances using mould fungus



29. The branch of biology which studies about the change in the characteristics of a species over several generations based on natural selection is:

A. Ecology B. Genetics C. Evolution D. Taxonomy

30. Why in most leaves the stomata are more abundant in the lower epidermis?

A. To reduce water loss

B. To get sunlight C. To synthesize their food effectively

D. To increase water loss

31. Which one of the following is a correct statement about the flower of a plant?

A. A complete flower has 3 main floral parts

B. A perfect flower has only stamen

C.A perfect flower has only pistil

D. A perfect flower has both stamen & pistil

32. Which property of water allows some organisms to exploit the surface of water as a habitat?

A. High specific heat capacity.

B. High latent heat of vaporization

C. High surface tension and cohesion

D. High density and freezing properties

33. Which of the following is responsible for making every amino acid unique?

A. Amino group B. Hydrogen C. Carboxyl group D. "R" group

34. One of the following is not the end result of mitosis:

A. Growth of eukaryotic organisms

B. Replacement of damaged cells

C. Production of two identical daughter cells

D. Production of four different daughter cells



35. Which one of the following is not a property of decomposers?

A. Digestion of food takes place outside their body.

B. They are involved in nutrient recycling.

C. Digestion of food takes place inside their body.

D. They are important in cleaning the environment.

36. Which of the following is the advantage of an electron microscope?

A. Selects suitable environment and needs optimum temperature

B. Has low magnification and resolving power than light microscope

C. Observes complicated structure inside cell organelles

D. Specimens are examined in vacuum

37. One of the following is correctly matched structure with their function:

A. Nucleus — Site of chemical reactions

B. Mitochondria — Site of production of energy

C. Ribosome — Store genetic material

D. Nerve cells — for synthesis of protein

38. Assume a plant cell contains 70% salt and 30% water and is then immersed into a hypotonic solution. What will be the shape of the cell?

A. burst B. shrinks C. flaccid D. plasmolysed

39. When Grade 12 students do an experiment on the unknown food sample by using Benedict's test and then observe an orange-red precipitate, what was the food item to be tested?

A. Reducing sugar C. Lipids

B. Non-reducing sugar D. Proteins

40. In which part of the digestive system that changes peptone into peptides? In-------

A. Large intestine B. Ileum C. Duodenum D. Stomach

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41. during inhalation:

A. Diaphragm muscle contracts and flattens.

B. Ribs move down and in.

C. Internal intercostal muscle contracts.

D. Pressure inside the lung is higher than outside.

42. The immunity passed from mother to child during breastfeeding is:

A. Natural active B. Natural passive C. Artificial active D. Artificial passive

43. Which of the following is a correct match of Kingdom with its example?

A. Kingdom Monera → Plasmodium falciparum

B. Kingdom Protista → Mycobacterium tuberculosis

C. Kingdom Fungi → Mosses and liverworts

D. Kingdom Animalia → Echinodermata and Mollusca

Question #44 is answered based on the following list:

1. Reptile 2. Arachnids 3. Birds 4. Amphibians 5. Crustacea 6. Fish

44. Which animal groups are invertebrates?

A. 1, 2, 5 and 6 C. 1, 3, 5 and 6

B. 2 and 5 D. 2, 5, 4 and 6

45. How the normal microbionta prevents pathogens from our body?.

A. decrease toxic compounds C. Supplying available nutrients

B. inactivate the adaptive immune system D. covering the binding site of the pathogens

46. The bacteria that decrease the amount of nitrate in the soil is?

A. nitrogen fixing bacteria C. nitrifying bacteria

B. ammonium bacteria D. denitrifying bacteria

47. Which of the following is not an application of biology in the environment?

A. Producing medicine
B. Producing bio gas
C. Producing bio fuel
D. solid waste treatment

48. Which one of the following is not an output of the light-dependent stage of photosynthesis?

A. ATP B. CO_2 C. Oxygen D. NADPH

49. Photosynthesis cannot take place in aquatic environment more than 1000 meter depth because of;

A. Only blue light penetrates this far
B. Too cold
C. no light can penetrate this far
D. Too hot

50. How many ATP are required to produce one hexose sugar molecule?

A. 12 ATP B. 6 ATP C. 18 ATP D. 16 ATP

51. Which one of the following enzymes catalyzes two substrates at different times?

A. PEP B. ATP synthase C. Rubisco D. Kinase

52. How many ATP molecules are produced from the oxidation of one pyruvate molecule?

A. 18 B. 36 C. 38 D. 15

53. In which stage of cellular respiration does decarboxylation take place?

A. Glycolysis & Krebs cycle
B. Krebs cycle & link reaction
C. ETC & Krebs cycle
D. ETC & link reaction

54. Which one of the following is not a correct statement about fermentation?

A. Alcohol fermentation has no intermediate electron acceptor
B. Alcohol fermentation produces only ethanol
C. Lactic acid fermentation has no intermediate electron acceptor
D. Lactic acid fermentation produces carbon dioxide



55. Which one of the following events occurs during the daytime in the CAM photosynthesis pathway?

A. Oxaloacetate is formed

C. The stomata open to take in CO_2

B. CO_2 enters the Calvin cycle

D. CO_2 reacts with malate

56. A biologist mixes certain gases in a flask and boils to produce organic compounds, then the biologist concludes life comes from organic compounds. Which theory of the origin of life supports this idea?

A. Special creation B. Spontaneous generation C. Eternal life D. Biochemical origin

57. The organisms that first colonized Earth are:

A. Achaea bacteria B. Eubacteria C. Eukaryotic D. Invertebrate

58. Which one of the following is not a postulate of Lamarck?

A. New need

C. Use and disuse organs

B. Natural selection

D. Inheritance of acquired characteristics

59. The evidence that supports evolution by comparing DNA & hemoglobin among organisms is:

A. Comparative embryology

C. Comparative biochemistry

B. Comparative paleontology

D. Comparative anatomy

60. If a certain a certain radioactive carbon substance is decomposed and has left 3.125 percent from its original mass ., What is the age of the fossil?

A. 28650 yrs B. 22920 yrs C. 17190yrs D. 5730yrs

61. In which era did the first flowering plants appear?

A. Cenozoic B. Mesozoic C. Paleozoic D. Precambrian

62. Based on the amino acid sequence which animal is distantly related to human beings?

A. lamprey B. Frog C. Macaque D. Dog

63. A fox with thick fur is selected in the environment, while a fox with thin fur is not selected. Which type of natural selection is this?

A. Stabilizing B. Bell-shaped C. Directional D. Disruptive

D 64. All are prezygotic reproductive isolation mechanisms except--------isolation.

A. Gametic ✓ B. Mechanical ✓ C. Behavioral ✓ D. Hybrid breakdown

A 65. Which one of the following is the correct sequence of fossils in the human family from ancient to recent?

A. Ardipithecus → Australopithecus → Homo habilis → Homo erectus ✓

B. Australopithecus → Homo habilis → Homo erectus → Australopithecus

C. Homo habilis → Homo erectus → Ardipithecus → Australopithecus

D. Homo erectus → Homo habilis → Ardipithecus → Australopithecus

A 66. The movement of genetic material into and out of a population is:

A. Gene flow B. Bottleneck effect C. Founder effect D. Genetic drift

B 67. The difference between the nervous system and the endocrine system is:

A. Nerves transmit chemical messages while hormones transmit electrical impulses

B. Hormones transmit chemical messages while nerves transmit electricalimpulse

C. Hormones response messages quickly than nerves

D. Both nerves and hormones respond slowly

A 68. Which of the following statements correctly placed?

A. afferent neurons carry impulse from different part of the body to the CNS ✓

B. effector neurons carry impulse from different body parts to the CNS

C. Inter neurons carry impulses from different body parts to the CNS

D. Afferent neurons carry impulse from the CNS to different part of the body



69. Which one of the following does not happen when an axon is at rest?

A. The axon membrane is polarized.

B. The outside membrane is positive.

C. The outside membrane of an axon is negative.

D. The inside membrane of an axon is negative.

70. A person under the increasing influence of alcohol first loses his ability to think clearly, second his ability to walk straight, third his ability to breathe normally. Based on this evidence, which is the corre order in which alcohol affects the central nervous system?

A. Cerebrum → Cerebellum → medulla

B. Cerebellum → medulla → cerebrum

C. Medulla → Cerebellum → Cerebrum

D. Cerebellum → Cerebrum → Medulla

71. An action that does not take place during sympathetic system of the body is:

A. Decrease heart beat

B. Increase breathing rate

C. Dilating the pupil

D. Stimulate glucose release

72. The layer of the epidermis that stops water loss and protect against entry by pathogens is:

A. Melanocytes B. Keratin C. Langerhans cells D. Collagen

73. The two parts of the eye that refract light rays are:

A. Retina & cornea

B. Cornea & lens

C. Aqueous humor & lens

D. Vitreous humor & pupil

74. What happens to the circular and radial muscles during dim light?

A. Radial muscles contract & circular muscles relax

B. Circular muscles contract & radial muscles relax

C. Both radial and circular muscles contract

D. Both circular and radial muscles relax

75. The correct sequence of middle ear bones from outside to inside is:

A. Stirrup → Malleus → Incus

B. Malleus → Stirrup → Incus

C. Malleus → Incus → Stirrup

D. Stirrup → Incus → Malleus

76. A hormone secreted from the posterior pituitary gland is:

A. Growth hormone

B. Follicle-stimulating hormone

C. Thyroid-stimulating hormone

D. Oxytocin

77. Which one of the following is true about glucagon?

A. Increases glucose level in the blood

B. Decreases glucose level in the blood

C. Changes glucose to glycogen

D. Stores glucose in the kidney

78. The molecule which is not found in the glomular filtrate is:

A. Glucose B. Protein C. Urea D. Urine

79. The greenhouse gas which is mostly found in fertilizer is;

A. Nitrous oxide B. Methane C. Carbon dioxide D. hydro fluoro carbon

80. Among the following options, identify the air pollutant that can cause acid rain and its effect respectivel

A. Carbon dioxide and methane – melting of ice caps and rising oceans

B. Sulphur dioxide and nitrogen dioxide – damaging plants and water

C. Perfluorocarbons – causing skin cancer and cataracts

D. Chlorofluorocarbons – increasing UV rays and respiratory disease

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MODEL-EXAM ONE Answer Sheet **Subject:** Biology

Allotted time: 1:40'

Name Hanna Ayalew **Sec.** 12th C. **No.** 24

76/80 x1

Direction: Blacken the circles given next to the numbers as per your answer's choice.

	A	B	C	D
1.	O	●	O	O
2.	O	●	O	O
3.	●	O	O	O
4.	O	O	●	O
5.	O	O	●	O
6.	O	O	●	O
7.	O	O	O	●
8.	O	●	O	O
9.	●	O	O	O
10.	O	●	O	O
11.	O	O	Ø	●
12.	O	●	O	O
13.	O	O	●	O
14.	●	O	O	O
15.	O	●	O	O
16.	O	●	O	O
17.	O	●	O	O
18.	O	O	O	●
19.	O	O	●	O
20.	●	O	O	O
21.	O	●	O	O
22.	O	●	O	O
23.	O	O	●	O
24.	●	O	O	O
25.	O	O	O	●
26.	O	●	O	O
27.	O	O	O	●
28.	O	●	O	O
29.	O	O	●	O
30.	●	O	O	O

	A	B	C	D
31.	O	O	O	●
32.	O	O	●	O
33.	O	O	O	●
34.	O	O	O	●
35.	O	O	●	O
36.	O	O	●	O
37.	O	●	O	O
38.	O	O	●	O
39.	●	O	O	O
40.	O	O	Ø	●
41.	●	O	O	O
42.	O	●	O	O
43.	O	O	O	●
44.	O	●	O	O
45.	O	O	O	●
46.	O	O	O	●
47.	●	O	O	O
48.	O	●	O	O
49.	O	O	●	O
50.	O	O	●	O
51.	O	O	●	O
52.	O	O	O	●
53.	O	●	O	O
54.	O	●	O	O
55.	O	●	O	O
56.	O	O	O	●
57.	●	O	O	O
58.	O	●	O	O
59.	O	O	●	O
60.	●	O	O	O

	A	B	C	D
61.	O	●	O	O
62.	●	O	O	O
63.	O	O	●	O
64.	O	O	O	●
65.	●	O	O	O
66.	●	O	O	O
67.	O	●	O	O
68.	●	O	O	O
69.	O	O	●	O
70.	●	O	O	O
71.	●	O	O	O
72.	O	Ø	●	O
73.	O	●	O	O
74.	●	O	O	O
75.	O	●	Ø	O
76.	O	O	O	●
77.	●	O	O	O
78.	O	●	O	O
79.	●	O	O	O
80.	O	●	O	O
81.	O	O	O	O
82.	O	O	O	O
83.	O	O	O	O
84.	O	O	O	O
85.	O	O	O	O
86.	O	O	O	O
87.	O	O	O	O
88.	O	O	O	O
89.	O	O	O	O
90.	O	O	O	O

	A	B	C	D
91.	O	O	O	O
92.	O	O	O	O
93.	O	O	O	O
94.	O	O	O	O
95.	O	O	O	O
96.	O	O	O	O
97.	O	O	O	O
98.	O	O	O	O
99.	O	O	O	O
100.	O	O	O	O
101.	O	O	O	O
102.	O	O	O	O
103.	O	O	O	O
104.	O	O	O	O
105.	O	O	O	O
106.	O	O	O	O
107.	O	O	O	O
108.	O	O	O	O
109.	O	O	O	O
110.	O	O	O	O

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Biology Model Examination for grade 12th.

Time allotted:- 2:00'

May. 2018

Name ---Hanna Ayalew---------- Sec. ------ No. -------

I.Choose the best answer from the given alternatives.

1. Who were Ethiopian renowned biologist that are pioneer to indigenous trees of Ethiopia ?

   A. Pr .Beyene Petros B .Pr Gebisa Ejeta C .Pr Legese Negash D. Pr T ilahun .Y

2. Parts of microscope which regulates the amount of light that enters into the stage is ?

   A .Diaphragm B .Mirror C .Stage clip D .Eye piece

3. Which of the following structures of a cell is correctly matched with its function ?

   A . vacuole a place where most chemical reactions takes place

   B .Rough endoplasmic reticulum vital for protein synthesis

   C .Mitochondria controls cell activities

   D .Ribosomes involved in lipid synthesis

4. Which of the following organelles is missing in prokaryotic cells ?

   A .Cytoplasm B .Cell membrane C .Nuclear membrane D .DNA

5. A grade 12 student demonstrated the effects of osmosis in plant cells using onion epidermal ells in distilled water for 30minutes then it observed under a microscope .The cells he /she observed became ?

   A .Burst B .Turgid C .Plasmolysed D .Unchanged

6. What is the key advantage of of sexual reproduction in the context of disease resistance ?

   A .It produces more individuals

   B .The offspring are identical to the parent

   C .Genetic variations creates a higher probability that some offspring will survive new diseases

   D .It allows for the rapid spread of favorable mutations

7. What is the function of a condenser?

   A .Magnifies the image C .Moves the stage

   B. Focuses light on the specimen D .Holds objectives

8. Which of these is incorrect classification combination ?

   A .Porifera ----- canal system C .Arthropoda ------ jointed appendages

   B .Echinodermata ---- flame cells D .Coelenterates ------ Nematocysts

9. A plant that has vascular tissue , has seeds and produces cones as reproductive structures is classified as ?

   A .Bryophyte B .Pteridophyta C .Gymnosperms D .Algae

10. Which of the following is an incorrect statement regarding the characteristics of chordates ?

    A .Dorsal nerve chord C .Pharyngeal gill silts

    B .Heart is dorsal in position D .Post anal tail is present at some stage

11. The respiratory structure book lungs is characteristics of which type of Arthropods ?

A .Crustaceans B .Insecta C .Arachnida D .Chilopoda

12. Which of the following is not a characteristics of the amniotic egg that allowed early reptiles to fully transition to terrestrial life ?

A .Amnion B .Allantios C .Chorion D .Larval stage

13. In a marine ecosystem , a pesticide accumulates in the tissue of organisms ,in which organism would you expect to find the highest concentration (bioaccumulation) ?

A .Zooplankton B .Phytoplankton C Os .prey (apex predator) D .Small fish

14. If the producers in a grass land ecosystems provide ,10 ,0000kJ of energy ,how much energy is available to the secondary consumers ?

A .100 KJ B .1000 KJ C .10 KJ D .1 KJ

15. Which organisms are critical for recycling nutrients ,acting as the final link in any food chains ?

A .Decomposers B .Apex predators C .Primary consumers D .Herbivorous

16. Which of the following is considered as a primary lymphoid organ ?

A .Spleen B .Thymus C .Lymph nodes D .Tonsils

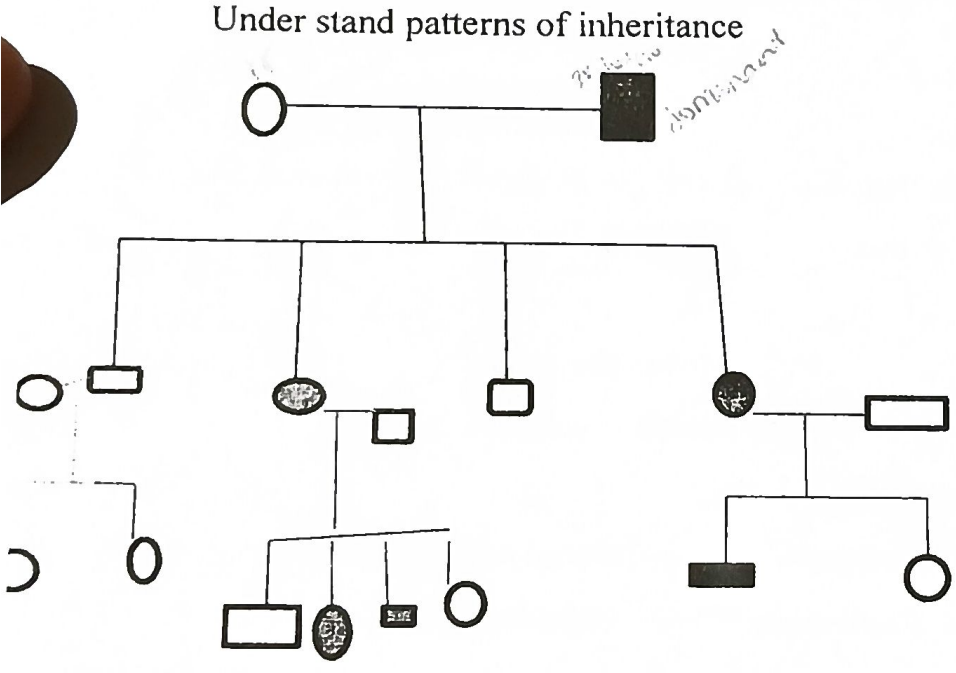
17. Which area do not have lymphatic vessels ?

A .Heart B .CNS C .Liver D . Dermis

18. The thoracic duct empties lymph in to which vein ?

A .Left sub clavian vein B .Right subclavian vein C .Jugular vein D .Superior venacava

Under stand patterns of inheritance



Key

Affected male

Affected female

Normal male

Normal female

19. The pedigree above shows the sex linked inheritance pattern of a gene this pattern of inheritance suggests the allele is?

A .X –linked dominant B .X –liked recessive C .Autosomal dominant D .Autosomal recessive

20. Which of the following correctly describes how an endothelium would respond to an increase in temperature?

A .Dilation of arteries near the surface of the skin

B .Erector muscles contract, causing hairs to stand up

C .Rapid contraction of skeletal muscles

D .Sweat glands releases less sweat

21. During the period of ventricular ejection in the cardiac cycle which of the following is true ?

A .Atrioventricular valves are closed and semilunar valves open

B .Atrioventricular valves are open and semilunar valves are close

C .Both Atrioventricular and semilunar valves are closed

D .Both atrio ventricular and semi lunar valves are open

22. Which factor stimulates the bone marrow to produce red blood cells in response to low oxygen levels ?

A .Thrombopoieton B .Interleukin 3 C .Erythropoietin D .Angiotensin

23. The most abundant cells of the blood are ?

A .Platelets B .Erythrocytes C .Leukocytes D .Granulocytes

24. In a diploid organism (2n =16),how many sister chromatids are there in a cell at the beginning of meiosis II (after cytokinesis of meiosis I)

A .8 B .32 C .64 D .16

25. Which of the following best describes the structural differences between mitotic metaphase and meiotic metaphase?

A .Mitotic metaphase aligns sister chromatids where as metaphase I aligns bivalents

B .Mitotic metaphase aligns homologous pairs ,metaphase I aligns individual chromosomes

C .Mitotic metaphase aligns bivalents ,metaphase I aligns sister chromatids

D .Both are identical in structure and function

26. Which condition is defined as an inability of the heart to pump sufficient blood to meet the body's needs ?

A . Sudden cardiac arrest B .Heart failure C .Aortic aneurisms D .Heart attack

27. What are the oval shaped masses of lymph tissue found in the neck ?

A .Cervical lymph nodes C .Axillary lymph nodes

B .Inguinal lymph nodes D .tonsils

28. Vascular bundles in a monocot stem described as closed because ?

A .They are surrounded by by sclernchymatous bundle sheath

B .They lack cambium between xylem and phloem

C .They are arranged in a closed ring

D Phloem is located in the center

29. For which of the following compounds does cholesterol not serve as a precursor ?

A .Bile pigments B .Bile salts C. Vitamin D D .Sex hormones

0. Which of the following statements is true about the primary structure of proteins ?

A .The helical structure of a protein

B .Sub unit structure of a protein

C .Three dimensional structure of a protein

D .The sequence of amino acids joined by a peptide bond

31. The parts of the large intestine ,starting from the part closest to the small intestine is ?

A .Caecum ,ascending ,transverse ,descending and sigmoid colon ,rectum and anal canal

B .Caecum , ascending ,transverse ,descending and sigmoid ,anal canal and rectum

C .Caecum ,descending ,transverse ,,ascending ,sigmoid rectum and anal canal

D. Ascending ,caecum ,transverse ,descending and sigmoid ,rectum and anal canal

32. A reduction in the activity of the entero kinase would directly inhibit the activation of ?

A. Lipase B. Amylase C .Pepsinogen to pepsin D .Trypsinogen to trypsin

33. Which hormone released in response to acidic chyme in the duodenum ,stimulates the pancreas to secrete bicarbonate rich fluids ?

A .Gastrin B .Secretin C .Motilin D .Cholesteokinin

34. In an experiment studying the toxicity of a new compound the researcher ensures that animal discomfort is minimized by using alternative non –invasive imaging techniques ,this best illustrates which principle ?

A .Replacement B .Reduction C .Refinement D .Replication

35. Portions of DNA molecules that do not contain the codes for proteins are called ?

A .Exons B .Introns C .Operons D .Mutagen

36. Which of the following is not true about DNA in prokaryotes ?

A .Present in the form of a compact structure called nucleoli

B .The coils are maintained by non –histone basic proteins

C .Found in cytoplasm in a super coiled condition

D .Packaged in nucleosomes along with histones

37. The one gene one polypeptide theory states that ?

A .The function of an individual gene is to dictate the production of a specific polypeptide

B .The synthesis of each gene is catalyzed by one specific enzyme

C. Each polypeptide catalyzes a specific reaction

D .The synthesis of each gene is catalyzed by one specific enzyme

38. Sickle cell anemia is caused ?

A .When valine is replaced by glutamic acid in beta poly peptide chain

B. When glutamic acid is replaced by valine in beta polypeptide chain

C .When valine is replaced by glutamic acid in alpha polypeptide chain

D .When glutamic acid is replaced by valine in alpha polypeptide chain

39. Which enzyme class is responsible for joining two molecules together using ATP ?

A .Hydrolases B. Ligases C. Lyases D .Oxidoreductases

40. Allosteric modulator influences enzyme activity by ?

A .Binding to the active site

B .Binding to the site other than the active site

C .Irreversibly denaturing enzyme

D .Changing the temperature of the reaction

41. Which of the following is responsible for maintaining the tertiary structure of an enzyme , making it highly sensitive to PH changes ?

A .Peptide bonds B .Ionic bonds C .Disulfide bonds D .Hydrophobic interactions

42. A base substitution mutation that changes a codon specifying an amino acid in to a stop codon is called ?

A .Missense mutation B .Nonsense mutation C .Silent mutation D .Frame shift mutation

43. For a sex linked recessive disorder ,if a woman who is a carrier marries a normal male ,what is the probability that their first son will be affected ?

A 0% B 25 % C .50 % D . 75 %

44. In crossing a homozygous recessive with a heterozygous ,what is the chance of getting an offspring with the homozygous recessive genotype ?

A .75 % B ,50 % C.25 % D . 0 %

45. Which type of inhibitor binds only the enzyme substrate complex and not to the free enzyme ?

A .Competitive B .Uncompetitive C .Non competitive D .Allosteric

46. The induced fit model of enzyme catalysis proposes that ?

A .The substrate changes shape to fit to the active site

B .The active site changes shape upon binding to the substrate

C .The enzyme and substrate are rigid structures

D .The enzyme remains rigid when the substrate binds

47. Which tool is most accurate for measuring exactly 15 ml of liquid ?

A .Beaker B .Elienymer flask C .Graduated cylinder D .Test tube

48. Why is solid water (ice) less dense than liquid water ?

A .Hydrogen bonds breaks when water freezes

B .Water molecules pack together in a solid state

C .A crystal lattice forms ,holding molecules further apart

D .The molecules become smaller

49. The ability of water to move up a plants xylem against gravity is primarily due to ?

A .High surface tension and low cohesion C .Cohesion and Adhesion

B .Adhesion and high latent heat D .Density and viscosity

50. Which structural muscle part is specialized for storing and releasing calcium ions for contraction

A .Troponin B .Myosin filaments C .Sarcoplasmic reticulum D .Actin

51. Which bone cell is responsible for bone resorption ?

A .Osteoclast B .Osteoblast C. Osteocyte D .Chondrocyte

52. Which type of joint allows movement in all directions ?

A .Hinge joint B .Ball and socket joint C .Pivot joint D .Saddle joint

53. What distinguishes operant conditioning from classical conditioning ?

A .Operant conditioning involves reflex
C .Operant conditioning is influenced by consequences.
B .Classical conditioning uses shaping
D .Classical conditioning strengthens voluntary actions

54. When analyzing a graph of enzyme activity a bell shaped curve indicates that the reaction rate is affected by ?

A .Substrate concentration
B .PH
C .Enzyme concentration
D .Inhibitors

55. A client using a diaphragm should be instructed to ?

A .Remove it immediately after intercourse
C .use it without spermicide for best results
B. Leave it in place for at least 6 hours after intercourse
D .Replace it every month

56. What is the function of placenta during pregnancy ?

A Produce estrogen
B .Secretes progesterone
C .Provide nutrients and oxygen to the fetus
D .B& C

57. Which of the following represents the correct sequence of stages in the sexual reproduction of fungi

A. Karyogamy ⟶ plasmogamy ⟶ meiosis
C. plasmogamy ⟶ karyogamy ⟶ meiosis
B. Meiosis ⟶ plasmogamy ⟶ karyogamy
D. karyogamy ⟶ meiosis ⟶ plasmogamy

58. Assume you are trying to demonstrate breathing movements using artificial lung and pulled down the rubber sheet the action of pulling represents which step of the actual breathing process ?

A .Exhalation and relaxation of the diaphragm
C .Inhalation and relaxation of diaphragm
B .Inhalation and contraction of diaphragm
D .none

59. The joint between the atlas and axis of the neck is called ?

A .Gliding
B .Condlyoid
C. Hinge
D .Pivot

60. What is the purpose of cervical mucus during ovulation ?

A .Facilitate sperm movement through the cervix
C .Maintain endometrial lining
B .Prevent sperm from entering the uterus
D. Block the fallopian tubes

61. Methanogenic bacteria are archaes that are ?

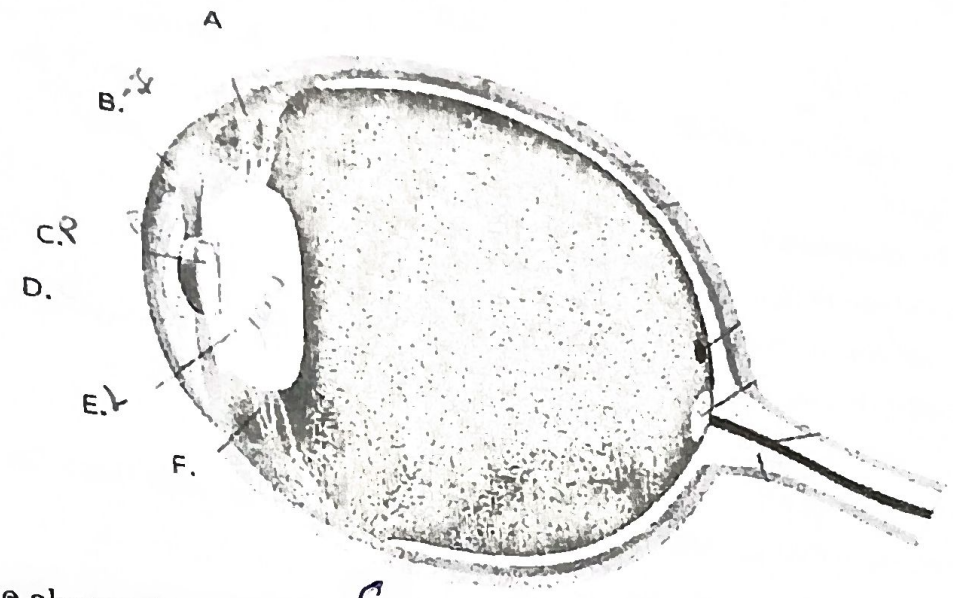
A .Strict aerobes that obtain energy through the synthesis of methane
B .Strict anaerobes that obtain energy through the synthesis of methane
C .Strict anaerobes that obtain energy through the consumption of methane
D .All

62. All except one is the characteristics of the lysogenic cycle ?

A .Many bacterial cells containing viral DNA are produced
B .Viral DNA are incorporated into the host genome
C .The viral genome replicates without destroying the host
D .A large number of phages is released at a time

63. The active ,feeding multiplying stage of most protozoans are ?

A .Cyst stage
B .Trophozite stage
C .Egg stage
D .All

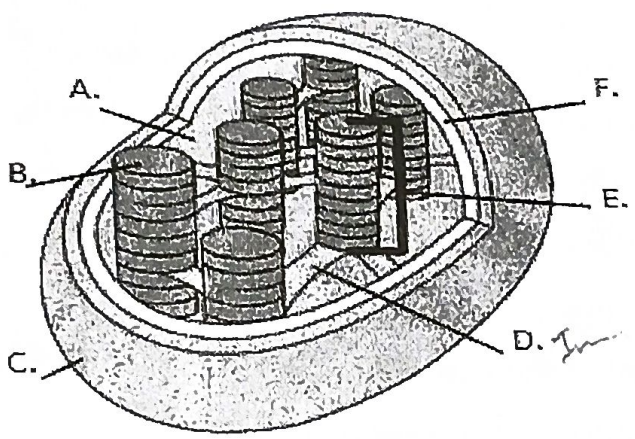


64. In the above diagram letter C Indicates ?

A .Iris B .Cornea C .Pupil D Lens

65. Based on the above eye diagram the part of the eye that controls the amount of light that passes through the pupil is indicated by letter?

A .B B .A C.D D .C



66. Based on the above diagram the part of the chloroplast on which dark reaction of photosynthesis takes place is indicated by letter ?

A .B B .A C .D D .C

67. What is the definition of a photo organography ?

A .An organism that uses light for energy and carbon dioxide for carbon

B .An organism that uses light for energy and organic compounds for carbon

C .An organism that uses in organic chemical for energy and carbon dioxide for carbon

D .An organism that uses organic chemicals for energy and organic compounds for carbon

68. A population ecologist observes a species that produces thousands offspring, provides no parental care ,and has a very high mortality rate among larvae ,but lower mortality among adult survivors, which type of survivor ship curve best depicts this population ?

A .Type I B .Type II C .Type III D .None

69. Which sexually transmitted infection is most likely to be associated chronic asymptomatic ,infection leading to high rate of transmission ?

A .Trichomoniasis B .Gonorrhea C .Chlamydia D .Cancroid

70. Which of the following is not true about archae ?

A .They can be found in extreme environments

B .They are most closely related to bacteria than eukarya

C .They include methanogens

D .Their r –RNA is different from bacteria

71. A 31 year old woman presents as she has noted an offensive fishy vaginal discharge ,she describes a grey watery discharge , what is the most likely diagnosis ?

A .Syphilis B .Candida C .Chlamydia D .Bacterial vaginosis

72. Bacterial endospores are most resistant to control methods because ?

A .They have a waxy cell wall C .They are metabolically inactive with protective coats

B .They contain multiple cell membrane D .none

73. The part of oviduct which is closest to the uterus is ?

A .infundibulum B .Fimbriae C .Isthmus D .Ampulla

74. Closest primates to man is ?

A Gorilla B .Rhesus monkey C .Orangutan D .Lemur

75. Which of the following is true about the importance of Lucy in resolving debates about human evolution ?

A .Big brain came before bipedalism C .Bipedalism came before big brain

B .Lucy had a brain size of 1.8 %of her body mass D .Lucy was partially an arboreal primate

76. What is the fate of 2 –phosphoglycolate formed in the chloroplast during the C_2 cycle ?

A .It is dephosphorylated and transported to the peroxisome as glycolate

B. It is exported directly to the mitochondria for further metabolism

C .It is immediately used to regenerate RUBP in the calvin cycle

D .It is directly converted to 3 PGA by carboxylase enzyme

77. Which of the following gases is primarily responsible for acid rain formation along with nitrogen oxides ?

A .CFCs B .SO_2 C. Methane D .Ozone

78. The energy released by ETC is used to pump H^+ in to which location ?

A .From cytoplasm to matrix C .From matrix to inter membrane space

B .From inter membrane space to matrix D .From inter membrane space to cristae

79. Which of the following components are directly involved in forming the light harvesting complex

A .Pigments B .Lipids C .Proteins D .A & C

80. Which layer of the skin containing no blood vessels and relies on the dermis for nutrient supply via diffusion ?

A .Reticular layer B .Epidermis C .Hypodermis D .Subcutaneous tissue

81. What is the function of the Langerhans cells found in the epidermis ?

A Detecting cold temperature

B .Acting as macrophages to fight infections

C .Producing keratin

D .Producing melanin

82. Semicircular canals are filled with ?

A Endolymph B .Perilymph C Mucous D .None

83. Which of the following structures is responsible for detecting linear acceleration ,such as riding in an elevator?

A .Semi circular canals B .Utriculus C .Sacculus D .Ampulla

84. Which structure connects the tongue in the floor of the mouth ?

A Papillae B .Frenulum C.Uvula D .Epiglottis

85. What is the main purpose of the cilia and mucus in the nasal cavity ?

A .To trap dust and microbes

B .To produce saliva

C .To increase air speed

D .To change the color of the nose

86. Which pigment is situated in the cone cells and is essential for bright vision ?

A .Retinal B .Opsin C .Iodopsin D .Rhodopsin

87. Which of the following is true regarding the dilator pupilae?

A .It is under parasympathetic control

B .It acts on the suspensory ligament

C. It constricts the pupil in bright light

D .It is under sympathetic control

88. Which nerve carries both auditory and vestibular system ?

A .Vagus nerve B .Facial nerve C .Vestibulocochlear nerve D .Glossopharyngeal nerve

89. Which of the following is a key difference between substrate level phosphorylation and oxidative phosphorylation ?

A .Substrate level involves direct transfer of a phosphate group ,while oxidative phosphorylation requires a proton gradient

B .Substrate level requires oxygen while oxidative does not

C .Substrate level phosphorylation happens in the matrix while oxidative phosphorylation in the inter membrane space

D Substrate level phosphorylation produces more ATP than oxidative phosphorylation

90. What is the primary mechanism by which aldosterone acts to regulate blood pressure and volume ?

A .It increases urea excretion

B. It increases Na^+ from the filtrate

C .It increases water reabsorption directly from the glomerulus

D .None

91. Which of the following is a classic example of a positive feedback mechanism in human physiology ?

A .Regulation of blood glucose by insulin and glucagon

B .Thermoregulation via sweat production and shivering

C .The surge of LH caused by high estrogen levels before ovulation

D .Regulation of calcium levels by parathyroid hormone

92. The set point in a negative feedback loop is best described as ?

A .The desired range or value for a homeostatic variable, managed by the integrator

B .The sensor that detects changes in the environment

C. The maximum physiological limit an organism can withstand

D .The thresh hold when reached shut down the effector

93. When blood glucose levels decreases how do alpha cells in the pancreas respond to maintain homeostasis?

A .By releasing glucagon to stimulate glycogen lysis and gluconeogenesis

B .By releasing insulin to inhibit glycogen lysis

C .By releasing glucagon to stimulate glucose uptake

D .By releasing insulin to trigger glycogenlysis

94. Which of the following hormones is not a counter regulatory hormone to insulin?

A .Aldosterone B .Cortisol C .Epinephrine D .Glucagon

95. In a nephron which part is virtually impermeable to water and play a crucial role in creating a dilute filtrate ?

A .Ascending limb of loop of henle

B .Descending limb of loop of Henle

C .Collecting duct

D .Proximal convoluted tubule

96. What is the molecular mechanism of peptide hormones /

A .Binding to intracellular receptors and altering gene transcription

B .Activating cell surface membrane receptors to initiate secondary messenger systems

C .Direct activation of mitochondrial enzymes

D .Diffusion directly through the nuclear membrane

97. What is the neuro transmitter responsible for pupillary constriction (sphincter action)

A .Nor epinephrine B .Acetylcholine C Adrenaline D .Dopamine

98. Which of the following best describes the function of nodes of Ranvier ?

A .Synthesis of neurotransmitters

B .Rapid generation of action potentials

C .Site of myelination

D Inhibition of nerve impulses

99. During the repolarization phase of an action potential which ion movement is primarily responsible for the rapid decrease in membrane potential ?

A .K^+ ion efflux B .Na^+ ion influx C . Na^+ ion efflux D . K^+ ion influx

100 . A cross between a chicken with white feathers and a chicken with black feathers ,this results in offspring with some black feathers and some white feathers ,what type of inheritance is this ?

A .Blended dominance B .Co dominance C .Incomplete dominance D .Complete dominance

101. Which technique is most commonly used for DNA amplification in medical diagnostics and forensics?

A. Western blotting

B. polymerization chain reaction

C. Enzyme linked immune sorbent assay

D. Flow cytometry

102. Which bacterium is widely used as bio pesticide in genetically modified crops?

A. Rhizobium legimunosarum

B. Bacillus thuringensis BT

C. Agrobacterium tumefactions

D. Agrobacterium tumefaciens

103. Which grain is the primary ingredient required for the production of Bourbon?

A. wheat B. Rye C. Barely D. corn

104. How do plants primarily absorb sulfur from the soli?

A. Elemental sulfur B. Hydrogen sulfide C. sulfate D. sulfuric acid

105. Which physical method of microbial control is primarily used to achieve complete sterilization?

A. pasteurization B. Autoclaving C. Boiling water for 10 minutes D. using 70% alcohol

106. Which of the following is a function of pasteurization

A. Killing pathogenic microorganisms ✓

B. Reducing the total microbial population

C. Increasing the shelf life of the product

D. All

107. Which component is a major structural feature of the gram positive bacterial cell wall

A. lipopoly saccharide B. outer membrane C. Teichoic acid D. porin proteins

Good Luck

Set by Begosew T

BLESSED G. MICHAEL CATHOLIC SECONDARY SCHOOL

MODEL-EXAM Answer Sheet **Subject:** Biology. 9 10

Allotted time: 2:10hr

Name Hanna Gyalew **Sec.** 12th C **No.** 21

Direction: Blacken the circles given next to the numbers as per your answer's choice.

	A	B	C	D
1.	O	O	Ø	●
2.	Ø	●	O	O
3.	O	●	O	O
4.	O	O	●	O
5.	O	●	O	O
6.	O	O	●	O
7.	O	●	O	O
8.	O	●	O	O
9.	O	O	●	O
10.	O	●	O	O
11.	O	●	Ø	O
12.	O	O	O	●
13.	O	O	●	O
14.	O	●	O	O
15.	●	O	O	O
16.	●	Ø	O	O
17.	O	●	O	O
18.	Ø	●	O	O
19.	●	O	O	O
20.	●	O	O	O
21.	●	O	O	O
22.	O	O	●	O
23.	O	●	O	O
24.	●	O	O	Ø
25.	●	O	O	O
26.	O	●	O	O
27.	●	O	O	O
28.	●	O	O	O
29.	Ø	●	O	O
30.	O	O	O	●

	A	B	C	D
31.	Ø	O	●	O
32.	O	O	O	●
33.	O	●	O	O
34.	●	O	O	O
35.	O	●	O	O
36.	O	O	O	●
37.	●	O	O	O
38.	O	Ø	O	●
39.	O	●	O	O
40.	O	●	O	O
41.	O	●	O	O
42.	O	●	O	O
43.	O	●	Ø	O
44.	O	●	O	O
45.	O	●	O	O
46.	O	●	O	O
47.	O	O	●	O
48.	O	O	●	O
49.	O	O	●	O
50.	O	O	●	O
51.	●	O	O	O
52.	O	●	O	O
53.	O	O	●	O
54.	●	Ø	O	O
55.	O	●	O	O
56.	O	O	O	●
57.	O	O	Ø	●
58.	O	●	O	O
59.	O	O	O	●
60.	Ø	●	O	O

	A	B	C	D
61.	O	●	O	O
62.	O	O	O	●
63.	O	●	O	O
64.	O	O	●	O
65.	●	O	O	O
66.	O	●	O	O
67.	O	●	O	O
68.	O	O	●	O
69.	O	●	O	O
70.	O	Ø	O	●
71.	O	O	O	●
72.	O	O	●	O
73.	O	O	●	O
74.	●	O	O	O
75.	O	O	●	O
76.	●	O	O	O
77.	O	●	O	O
78.	O	O	●	O
79.	O	O	O	●
80.	O	●	O	O
81.	O	●	O	O
82.	Ø	O	O	●
83.	●	O	Ø	O
84.	O	●	O	O
85.	●	O	O	O
86.	O	O	●	O
87.	O	O	O	●
88.	O	O	●	O
89.	●	O	O	O
90.	O	●	O	O

	A	B	C	D
91.	O	O	●	O
92.	Ø	O	●	O
93.	●	O	O	O
94.	●	O	O	O
95.	●	O	O	O
96.	O	●	O	O
97.	O	●	O	O
98.	O	●	O	O
99.	●	O	O	O
100.	O	●	O	O
101.	O	●	O	O
102.	O	●	O	O
103.	O	O	O	●
104.	O	O	●	O
105.	O	●	O	O
106.	O	O	O	●
107.	O	O	●	O
108.	O	O	O	O
109.	O	O	O	O
110.	O	O	O	O
111.	O	O	O	O
112.	O	O	O	O
113.	O	O	O	O
114.	O	O	O	O
115.	O	O	O	O